

Lesson Summary

The Secret of Codey Rocky

Goal of the Lesson

In this lesson, we will be introduced to the Codey Rocky robot and the basics of programming. We will explore what a program is, how robots like Codey Rocky can follow instructions, and how to upload simple programs using mBlock 5. The goal is to understand how we can use code to bring robots to life and make them perform various tasks.

Key Concepts and Steps:

What is Codey Rocky?

- A small educational robot that can display animations, avoid obstacles, and more.

What is a Program?

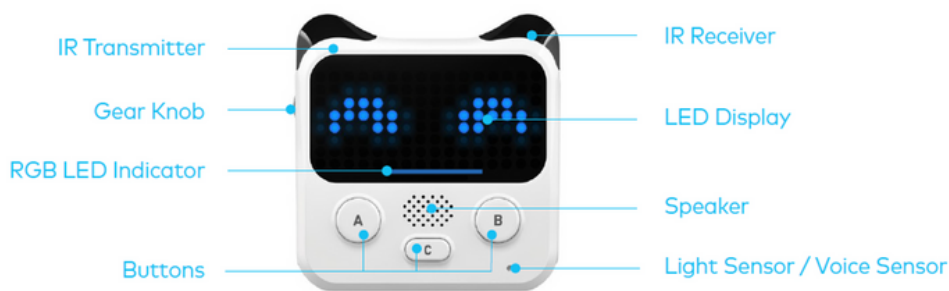
- A program is an artificial language that we use to tell robots what to do.

What is the structure Codey Rocky?

- It is composed of two main detachable parts:
 - The top part, Codey, that handles thinking and sensing.
 - The bottom part, Rocky, that handles moving and interacting with the physical world

What are the features of Codey?

- Codey contains all the electronics, sensors, and programmable components, including:
 - LED screen, Buttons, Gyroscope, Light sensor, Speaker, Infrared sensor, USB port, Bluetooth and Wi-Fi, and a Microcontroller.



What are the features of Rocky?

- Rocky enables physical movement and interaction, including:
 - Motors and wheels, Line-following sensors, Ports to connect external components

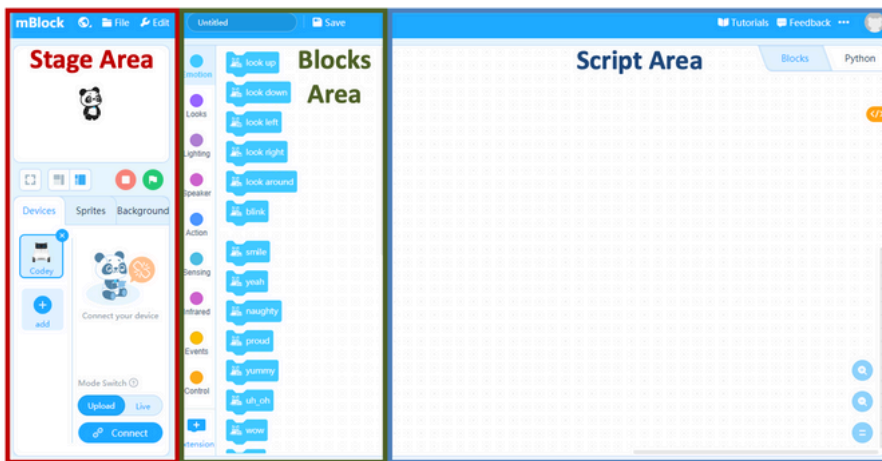


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What is mBlock 5?

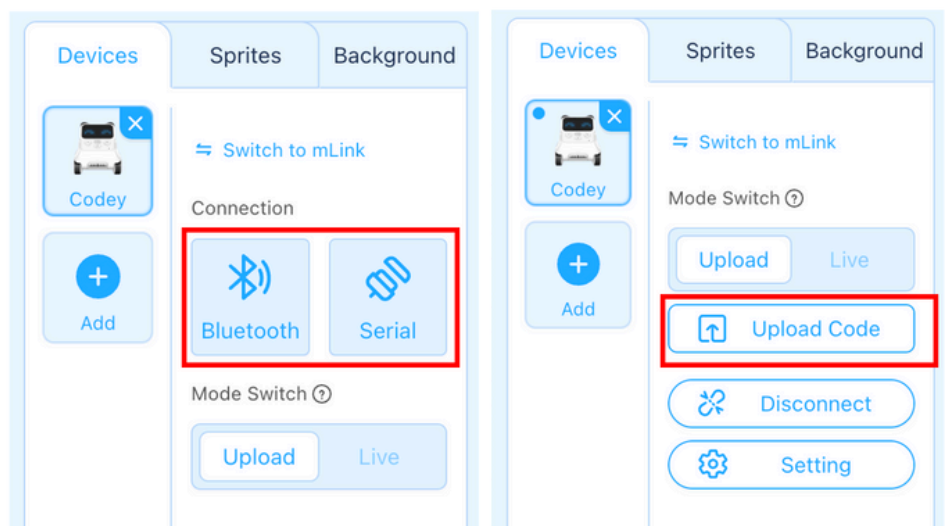
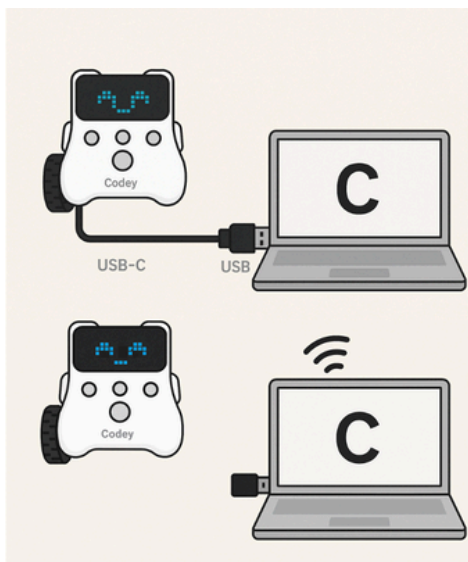
- A graphical programming environment used to create and upload code to Codey Rocky.



- The Stage Area: Shows the actions of your sprites and devices. You can connect your robot, switch modes, and preview results.
- The Blocks area: Contains all programming blocks, organized by categories like Motion, Events, and Control.
- The Script area: The workspace where you drag and snap blocks together to build your program

What are the steps to upload a program to Codey Rocky?

- Connect Codey Rocky to your computer via the USB cable or Bluetooth dongle.
- Open mBlock 5 and create a simple program.
- Choose the connection type: Bluetooth or Serial
- Click on the “Upload Code” button on the Mblock editor.
- Observe how Codey Rocky executes the instructions.



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Coding our project:

In this lesson, we created a simple program to control how Codey Rocky moves when different buttons are pressed:

- Program 1: When button A is pressed, Codey Rocky moves forward at 50% power for 5 seconds.
- Program 2: When button B is pressed, Codey Rocky moves backward at 50% power for 1 second.
- Program 3: When button C is pressed, Codey Rocky turns right at 50% power for 1 second.

